Yuchen You

East 20 Room 606, 800 Dongchuan Rd., Shanghai, China

Email : yuchenxr@umich.edu

Alt : wesley_you@sjtu.edu.cn

Education

- **University of Maryland** — 2026 – 2031 (Expected)
  PhD in Computer Science (Reliable Agentic Networked System) — *College Park, MD, US*
  - **Advisor**: *Prof. Zaoxing (Alan) Liu*
- **University of Michigan** — 2024 – 2026
  B.E. in Computer Science — *Ann Arbor, MI, US*
  - **Advisor**: *Prof. Ryan (Peng) Huang, Prof. Xiaonan (Sean) Huang*
  - **Honors**: Cheng-Family Scholarship (Top 2%), Dean's List, Honors List
- **Shanghai Jiao Tong University** — 2022 – 2026
  B.E. in Mechanical Engineering — *Shanghai, China*
  - **Advisor**: *Prof. Yutong Ban*
  - **Honors**: John Wu & Jane Sun Sunshine Scholarship (Top 6%), SJTU Excellence Scholarship (Top 3%)

Research Experience

- **Agentic Distributed System Overload Mitigation** — May 2024 – Present
  Order Lab, University of Michigan; Advisor: Prof. Ryan (Peng) Huang — *Ann Arbor, MI, US*
  - **Closed-loop Control Plane**: Engineered a **closed-loop control plane** for **ZooKeeper** clusters to mitigate overload failure, integrating **Prometheus** telemetry, **HAProxy, tc** for traffic shaping.
  - **Mitigation/RCA action library**: Implemented a library of **atomic mitigation actions**—including network throttling (**tcconfig**), dynamic load balancing (**HAProxy Runtime API**), **JVM** heap tuning, and disk I/O control (**fsync**)—packaged as auditable scripts that any controller can invoke safely.
  - **Load Injection & Evaluator**: Built a custom load generator (zkbench) that supports **multi-stage traffic** (warmup → spike → cooldown) to stress-test resilience, and developed a closed-loop evaluator that correlates injection rates, SLO violations, and goodput for different control strategies.
  - **Agentic Controller**: Prototyped an **Claude Code styled controller** (GPT-4.1 with **Model Context Protocol** tools) that consumes metrics and log summaries to do root cause analysis, select mitigation actions, enabling comparison between static policies and agentic reasoning on complex failure patterns. Tuned the context window usage for Models with log processor to cut down number of cycles.
- **SoftRobot Electronic Control** — Sept. 2024 – Present
  HDR Lab, University of Michigan; Advisor: Prof. Xiaonan (Sean) Huang — *Ann Arbor, MI, US*
  - **SOTA Soft Robot Trajectory Following Accuracy**: Reached **sub-mm** level trajectory following accuracy, and with flexible **Plug-and-Play Modular Design**.
  - **Real-time embedded control stack**: Architected a **real-time control stack** across **STM32** (low-level actuation) and **Orange Pi** (planning); designed custom PCBs and implemented **multi-threaded task isolation** to decouple high-frequency **CAN/I^{2}C** telemetry from planning and logging.
  - **Control algorithms and kinematics**: Implemented feedback control and a **Piecewise Constant Curvature** kinematic solver with Jacobian-based IK to handle high-DoF redundancy in real time, supporting smooth and repeatable soft-arm motion.
  - **Validation and Recognition**: Collaborated with and funded by **General Motors**; work recognized with an **ICRA 2025 Best Poster Award** and spotlight talks at **RoboSoft 2025** and **ICON 2025**.

Publications

- **Origami Inspired Soft Robotic Arm: A Modular Platform for Manipulation** — May 2025
  IEEE ICRA Workshop, Atlanta, USA — *Best Poster Award*
  - **Authors**: Jiyang Wang, **Yuchen You**, Xinqi Zhang, Haobo Fang, Jiaqi Wang, Xiaonan Huang
  - **Award**: Gave a **spotlight talk**; poster received the **Best Poster Award** at the IEEE ICRA 2025 Workshop.

Patent

- **U.S. Patent, 2025**: J. Wang, Y. You, X. Xhang, H. Fang, J. Wang, and X. Huang, "Lightweight, Proprioceptive, Origami-Inspired Soft Robotic Arm for High Payload, Low-Cost Reconfigurable Manipulation," (Pending), 2025.

Selected Projects

- **Search Engine "Crawl Me Maybe" in C++** Jan. 2026 – Apr. 2026
  EECS 498 (System Design for Search Engine) *Instructor: Nicole Hamilton, University of Michigan*
  - **Product Demo**: Build a complete search engine in team of 4, crawled **210M websites** within 2 days, indexed into **1TB invert index**, average query time **200ms**, p99 tail latency reaches **4s**, distributed across 20 GCP servers.
  - **Invert Index Read-Write Storage System**: Hierarchical design of storage system, utilizing **mmap** for disk-I/O, used **UTF-8 + VByte** encoding format and used **String-Pool** for variable length string storage. Implemented a **Recursive Skip-List** tree structrue and **Position List Cursor** for fast posting list walk. **High concurrency multi-threading** design and **Zero-Copy data path** to make index writing finally **Compute-Bound**.
  - **Constraint Solver Design**: Contributed a recursive constraint solver tree that accepts from the query compiler's AST tree and then do submatching and merging. Support **AND/OR/Phrase/NOT** logics.
  - **Ranker Design**: Contribute a benchmark with ˜100 hot query terms on Google/Bing and tune the dynamic ranker weights. Including BM25 tf;idf algorithm, domain name matching, last path matching, etc.
- **CUDA Proxy Player** Aug. 2025 – Dec. 2025
  CSE 582 (Advanced Operating Systems) – CUDA runtime optimization for LLM inference *University of Michigan*
  - **Multi-path CUDA runtime**: Designed a multi-path CUDA execution proxy (baseline / graph / persistent) that routes requests by batch size and micro-op shape to reduce CPU launch overhead on launch-bounded LLM inference workloads.
  - **Graph & persistent execution paths**: Implemented CUDA Graph capture and caching plus persistent worker kernels with device queues and shape bucketing, achieving up to **1.5–3**× lower end-to-end latency on repetitive, launch-bounded workloads.
  - **Microbenchmarks and instrumentation**: Designed microbenchmarks and runtime instrumentation (path statistics, latency, graph hit rate) to evaluate performance across MoE-like forward passes and routing strategies.
- **Simulated Distributed System in Golang** Aug. 2025 – Dec. 2025
  EECS 491 (Introduction to Distributed System)
  - **Primary–backup KV store**: Implemented a primary–backup key–value store in Go with lexical confinement and request concurrency, designing the protocol to provide linearizable GET/PUT/APPEND operations.
  - **Paxos-based replicated state machine**: Designed a three-layer Paxos replicated state machine and safely optimized the protocol with accept-phase skipping to reduce normal-case latency.
  - **Sharded KV & reconfiguration**: Implemented sharded key placement and dynamic reconfiguration using Paxos-based configuration management, debugging consistency and recovery under concurrent shard moves.
- **Simulated Basic Operating System** Jan. 2025 – Apr. 2025
  EECS 482 (Introduction to Operating System) Lecture Project
  - **User-level threading library**: Built a lightweight user-level multicore threading library (context switching, Mesa monitors, synchronization primitives) and designed scheduling to handle concurrent workloads without deadlock.
  - **Pager & virtual memory**: Implemented a minimal pager and MMU layer managing page tables and dirty/reference bits, designing a clock-based eviction policy and copy-on-write support for fork/mmap.
  - **Network file system & kernel tracing**: Implemented a Unix-style network file system with synchronized inode operations and extended Linux `ptrace` with selective memory snapshot/restore to support efficient debugging.
- **Unix Network Stack Simulation** Jan. 2025 – Apr. 2025
  EECS 489: Introduction to Computer Networks
  - **Mininet-based experiments**: Designed Mininet topologies and C++ socket experiments to measure RTT/throughput and reproduce bufferbloat, observing how queue sizes and AQM impact latency.
  - **Video proxy & transport**: Implemented a video proxy with basic load balancing and adaptive streaming, and built a TCP-like reliable transport over UDP plus an L3 router with ARP/ICMP handling.
  - **SDN controller**: Implemented a POX-based SDN controller that assigns flows to QoS queues, designing rules to mitigate bufferbloat for latency-sensitive traffic.
- **Digital Forensics** Jan. 2025 – Apr. 2025
  EECS 388 (Introduction to Computer Security) Lecture Project
  - **Cryptanalysis & Cracking**: length-extension, padding-oracle; John the Ripper (PDF/ODT), Hydra (SSH).
  - **Web Exploitation**: auth bypass via XSS/SQLi/CSRF.
  - **Binary Exploitation**: ROP/NOP-sled against DEP/ASLR.

- **Reverse Engineering**: Ghidra decompilation and PWNing.
- **Protocols**: TLS 1.3 handshake; Google-style TOTP.

- **Auto Sentry Robot Control** 2023 – 2024
  Chinese Univ. National Robot Competition – Robomaster Championship
  - **Autonomous control**: Autonomous decision making and engagement with dual gimbals and 4-wheel chassis on STM32-F407.
  - **Circuit & control**: Lead circuit design; dual-gimbal control stabilization; high-speed 4-wheel chassis response.
  - **Perception & pose**: Developed CAN/UART pipelines for CV and LiDAR data; implemented IMU-based absolute-pose control.

Skills

- **Programming**: C/C++, Golang, Java, Rust, Python, Bash; Git; CMake, Makefile, Maven, uv, cargo
- **Systems**: Arch/Ubuntu Linux; concurrency (boost locks); MMU/paging; POSIX sockets (select/poll)
- **Networking**: tc(config); HAProxy; Mininet, POX; TCP (GBN/SR), L3 routing
- **Distributed**: Docker (Compose), Kubernetes; ZooKeeper, HDFS; Prometheus+Grafana (JMX Exporter)
- **Security**: Wireshark, Ghidra, John the Ripper, Hydra, sqlmap, Autopsy, Stegseek; ROP chains
- **ML**: PyTorch, CoT, MCP, Qwen-VLM Tuning
- **Databases**: SQLite3 (sql-injection), Oracle(SQL*Plus)
- **Robotics**: STM32, FreeRTOS; CAN/I^2C; Flexiv RDK; PID/dynamics; C++/Rust firmware, MATLAB
- **Other**: JavaScript, HTML, Markdown, LaTeX; Neovim, SSH, tmux, GDB/LLDB

Honors & Awards

- **ICRA Best Poster Award (May 2025)**: Presented by IEEE Robotics and Automation Society.
- **Cheng–Family Scholarship (Jun. 2024)**: (Top 2%).
- **RoboMaster University Championship (May 2024)**: (Eastern Region Champion).
- **RoboMaster University League (Apr. 2024)**: (Shanghai Division Champion).
- **University Physics Competition (Nov. 2023)**: (Silver Prize).
- **SJTU Excellence Scholarship, Level B (Dec. 2023)**: (Top 3%).
- **John Wu & Jane Sun Sunshine Scholarship (Nov. 2023)**: (Top 6%).
- **RoboMaster University Championship (Aug. 2023)**: (National Champion).
- **SJTU Social Practice, Third Prize (Aug. 2023)**:

Teaching Experience

- **Teaching Assistant (2024)**: **Teaching Assistant** at Shanghai Jiao Tong University, ENGR 1000J (Introduction to Software Engineering).

Extra Curriculars

- **Undergraduate Research Assistant (May 2025 – Ongoing)**: Undergraduate Research Assistant at University of Michigan CSE Department, MI, USA.
- **ICRA Volunteer (2025)**: **Volunteer** at IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, May 2025.
- **UM-SJTU Joint Institute Youth Volunteer Team (2023)**: UM-SJTU Joint Institute Youth Volunteer Team member (Shanghai, China).
- **Old Friends Youth Team (2023)**: Old Friends Youth Team, Shanghai, Facilitated intergenerational communication activities.

Personal Infos

- **Language**: English (TOEFL 107/120), Chinese (Native).
- **Hobbies**: Badminton, Rubik's Cube, Linux Customization (Arch Linux + Hyprland + NeoVim + Claude Code).